

VGA-113-110

Dosing regulator, without differential pressure regulator

Installation and operating instructions



Other languages

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GRUNDFOS 

English (GB) Installation and operating instructions

Original installation and operating instructions

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Warning

These operating instructions are also available on www.Grundfosalldos.com.



Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Of general interest

1.1 Structure of the documentation

The Grundfos Alldos device VGA-113 are a state-of-the-art solution, which complies with recognised safety regulations. Conformity with applicable standards, directives and laws has been verified.

Nevertheless, certain risks which cannot be prevented by the manufacturer are associated with the use of the system.

Purpose of this manual:

- Inform users of optimum use.
- Warn users of possible residual risks when using correctly and identify measures that should be taken to avoid damage.
- Caution users against obvious misuse or inappropriate use and inform them of the necessary care that must be taken when operating the system.

1.2 About this manual

This manual contains the following standardised safety instructions about possible residual risks:



Warning

If these safety instructions are not observed, it may result in personal injury!

Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment!

Note

Notes or instructions that make the job easier and ensure safe operation.

Information about possible residual risks is provided:

- On warning signs displayed in the installation location.
- At the start of each section in this manual.
- Directly before any operating procedures that could involve residual risks.

1.3 User/target groups

Users are persons who are responsible for operating and monitoring the device VGA-113 at the installation location. The system may only be operated by trained and qualified personnel. Personnel must have appropriate technical knowledge and be familiar with the basic principles of measuring and control technology.

1.3.1 Responsibilities of the users

The users' responsibilities:

- Read this manual before operating the VGA-113.
- Be trained by qualified personnel from Grundfos Alldos in the operation of the system.
- Observe the recognised regulations governing safety in the workplace and accident prevention.
- Wear appropriate protective clothing in accordance with national regulations for the prevention of accidents when operating the system and handling chemicals (German GUV-V D05).

1.4 Responsibilities of the operator

The owner of the building or operator of the VGA-113 is responsible for the following:

- Consider this manual to be part of the product and ensure that it is kept clearly accessible in the immediate vicinity of the system for the entire service life of the system.
- Meet the installation requirements specified by the manufacturer (required water connections and fittings, environmental conditions, electrical connection, protective pipe for dosing line if necessary, audible or optical warning device for alarm messages if necessary).
- Ensure that water lines and fixings are regularly checked, serviced and maintained.
- Obtain official approval for storing chemicals, if necessary.
- Train users in the operation of the system.
- Ensure that the regulations for the prevention of accidents are observed in the installation location (German GUV-V D05 regulation for the prevention of accidents, "Chlorination of Water" dated January 1997).
- Provide all users and service personnel with protective clothing in accordance with GUV-V D05 (face mask, gloves, protective apron).

1.5 Maintenance and service personnel

The system may only be maintained and serviced by authorised service personnel from Grundfos Alldos.

1.6 Correct usage

The Grundfos Alldos VGA-113 may be used for dosing chlorine, Cl_2 , as being described in this manual.

1.7 Inappropriate usage

Applications other than those listed in section [1.6 Correct usage](#) are considered not to be in accordance with the intended use and are not permitted. The manufacturer, Grundfos Alldos, accepts no liability for any damage resulting from incorrect use.

The system comprises state-of-the-art components and has undergone safety-related testing.

Warning



Unauthorised structural modifications to the system may result in serious damage to equipment and personal injury.

It is forbidden to open, modify, change the structure of, bridge, remove, bypass or disable components, especially safety equipment.

2. Handling chlorine

2.1 Physical and chemical data

Under normal conditions of pressure and temperature, chlorine is a yellowish green gas with a pungent odour. It exists as diatomic molecule Cl_2 .

It is not flammable, but can under certain circumstances promote the flammability of metals, hydrocarbons etc.

Atomic weight	35,457
Molecular weight Cl_2	70,941
Density (liquid)	1,57 g/cm ³ at -34,05 °C
Density (gaseous)	3,214 g/l at 0 °C, 1 bar
1 l liquid chlorine at 0 °C	corresponds to 457 l (0,457 m ³) gaseous chlorine
1 kg liquid chlorine at 0 °C	corresponds to 311 l (0,311 m ³) gaseous chlorine
Specific gravity	2,486 (specific gravity of air: 1)
Boiling point	- 34,05 °C (1 bar)
Melting point	- 100,98 °C
Evaporation heat	269 kJ/kg (at 0 °C)
Heat conductivity	0,527 kJ/m ² h (liquid chlorine)
Degree of purity acc. to DIN 19607	99,5 %
TLV (Threshold Limit Value)	1,5 mg/m3 (0,5 Vol.-ppm)

Vapour pressure curve of chlorine

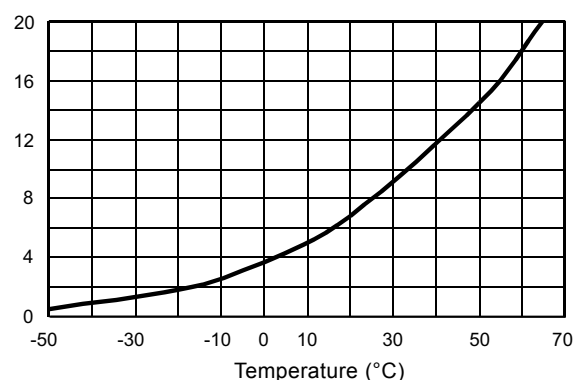


Fig. 1 Vapour pressure curve of chlorine

Solubility of chlorine gas in water

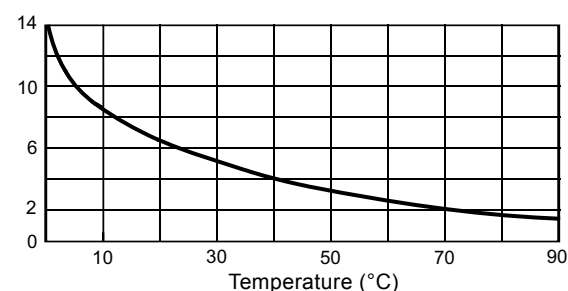


Fig. 2 Solubility of chlorine gas in water

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2.2 Safety advice for handling chlorine

2.2.1 Risks to health

Chlorine gas is toxic, more than 50 Vol.-% in the indoor air mean an acute danger to life.



Warning

R 23: Toxic by inhalation.

R 36/37/38: Irritating to eyes, respiratory system and skin.

Hazards of chlorine gas

- Irritating to eyes, respiratory system and skin.
- Causes whooping cough.
- Causes causticization of skin and respiratory system.
- Lethal by lung edema in case of long influence or high concentration.
- Slight paralyzing effect to the central nervous system.

Hazards of liquid chlorine

- Causes causticization of skin.
- Causes reddening and formation of bubbles.

2.2.2 Personal safety equipment

The operating authority of a chlorination plant has to provide for the operation personnel:

For each person

Respiratory equipment (full-sight gas mask)

- with an effective chlorine filter (B2P3), colour mark: grey with a white ring
- at least 1 spare filter per gas mask
- personally fitted (perfectly tight)
- labelled by name

Only plants with chlorine drums

- at least 2 protective suits with compressed air respirators

Storage of the safety equipment

- outside the chlorine rooms
- well visible
- easily available at any time
- protected from dust and moisture

Caution Further obligations of the operating authority

- Introducing the operation personnel to handle the safety equipment
- Carrying out exercises (at least half-yearly)
- Regular replacement of the gas mask filters
 - after the expiry of the date of durability
 - at least 6 months after opening (mark the opening date on the filter)
 - after contact with chlorine
 - Observe employing prohibition according to § 14 ArbStoffV (in Germany) resp. according to local laws!

2.2.3 Rules of conduct

- Change of chlorine containers only with gas mask.
- Entering contaminated rooms only with protective suit and compressed air respirator.
- In case of flight wear gas mask, if possible. Observe wind direction!
- Eating, drinking and storing food is prohibited in chlorine rooms.

2.2.4 First aid in case of accidents

First aid after having inhaled chlorine

- Keep calm.
- Remove injured persons from the dangerous area.
 - Helpers must pay attention to personal protection!
- Immediately remove contaminated clothes.
- Calm down injured persons and keep them warm with blankets.
- Supply fresh air; use oxygen respirator (alternately with inhaling steam), if possible.
 - No mouth-to-mouth resuscitation!
- Fast and gentle transport to hospital
 - lying
 - sitting in case of difficulty in breathing
 - state chlorine causticization as the cause.

First aid after causticization of the skin

- Keep calm.
- Remove contaminated clothes.
- Rinse skin with plenty of water
- Bandage the wound germ-freely
- Seek medical aid.
 - State chlorine causticization as the cause.

First aid after causticization of the eyes

- Keep calm.
- Rinse causticized eyes with plenty of water while the person is lying.
 - Protect healthy eye, if necessary.
 - Spread eyelids widely, let the eye move to all sides.
- Seek ophthalmologist.
 - State chlorine causticization as the cause.

First aid after internal causticization

- Keep calm.
- Drink water in short sips.
 - If possible, take medical charcoal.
- Seek medical aid.
 - state chlorine causticization as the cause.

2.2.5 Transport and storage of chlorine



Warning

Handling of chlorine containers only by experienced, practised personnel!

Basic rules for transport and storage of chlorine

- Treat containers carefully, do not throw!
- Protect containers from turning over or rolling away!
- Protect containers from direct sun rays and temperatures over 50 °C!
- Transport of containers only with valve protection nut and protection cap.



Warning

This rules are valid for both full and empty containers, as empty containers still contain rests of chlorine and therefore are under pressure.

Valid regulations

- Regulations for accident prevention "Chlorination of water" (GUV-V D5) with process instructions
- Regulations concerning places of work (ArbStättV))
- Technical rules for gases 280, 310 und 330



Warning

Strictly observe local law and regulations for handling, transport and storage of chlorine.

2.2.6 Pressure vessels and mountings

Chlorine is being offered in two container designs:

- Steel cylinders containing 50 kg or 65 kg, equipped with one valve for
 - the withdrawal of gaseous chlorine from the upright standing cylinder

Variant of chlorine cylinder

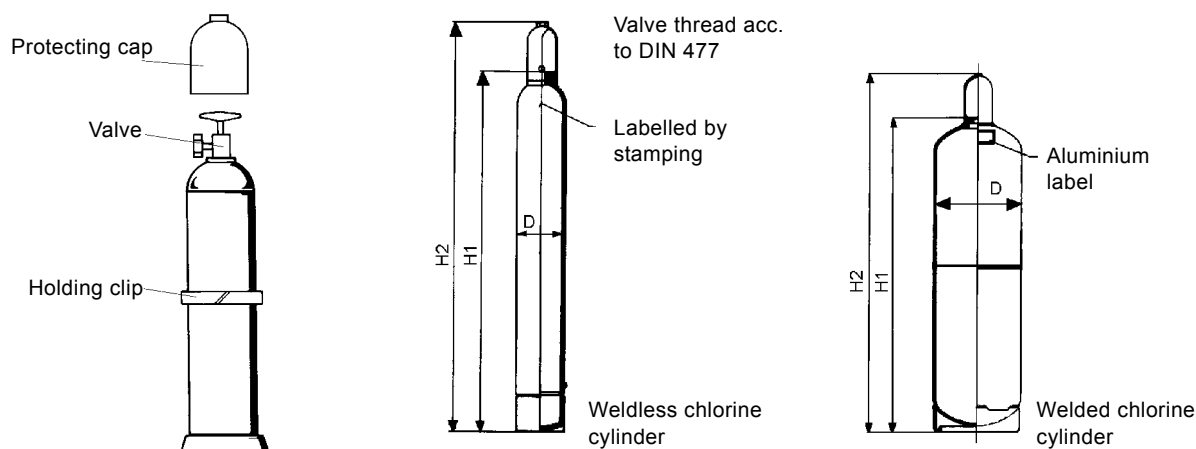


Fig. 3 Variant of chlorine cylinder

- Steel drums containing 500 kg or 1000 kg, equipped with
 - one valve for the withdrawal of gaseous chlorine
 - one valve for the withdrawal of liquid chlorine

Riser pipe (for the withdrawal of gaseous chlorine)

Tyre of roller

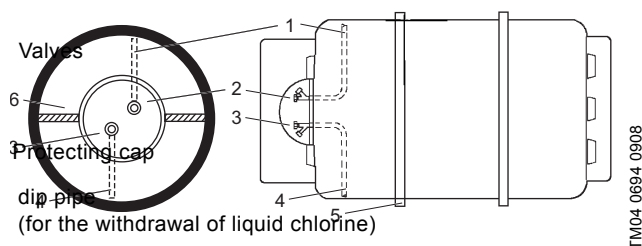


Fig. 4 Chlorine drum

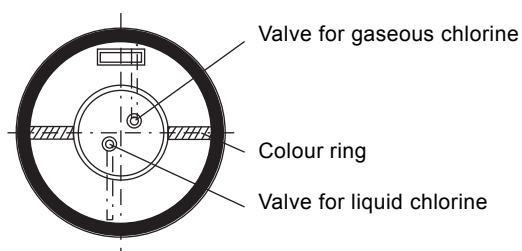


Fig. 5 Valve position of chlorine drum

Due to safety precautions, chlorine containers are only filled up to 95 % of their capacity.



Warning

Observe safety precautions for chlorine containers

- Kind of gas, weight, owner, producing date and date of the last testing have to be noted clearly on the container. Chlorine containers are marked by grey colour.
- No changes or repair by the user!
- Never open container valves by force. Stuck valve spindels can be loosened by wrapping a shred with warm water around the valve.
 - Never use an open flame!
 - Never use wrench lengthening!
 - Return containers with stuck valves to the manufacturer.
- Observe safety precautions and the manuals of the manufacturer!

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2.2.7 Chlorine extraction

Before the extraction

- The chlorine containers must be stored at least for 8 hours in the container room so that the content can adapt the ambient temperature.



Warning

Chlorine containers must never have a higher temperature than other parts of the plant. Danger of liquefaction and possible chlorine break-out!

- Turn chlorine drums on the support until the dip pipe and the riser pipe are placed vertically (observe the markings on the drum).
- Check tightness.

Connection

- Protect containers from turning over or rolling away!
- Dry the piping and the withdrawal system with dry nitrogen or dry air.
- No foreign matter must get into the plant.
- Mount new gaskets to the connection line.
- Connect the container.
- Slowly open the container valve.

Withdrawal of gaseous chlorine

At 15 °C about 1 % (DIN19607) of the content can be withdrawn per hour. In case of higher withdrawal danger of malfunction by the formation of ice on containers and piping!

Caution

Container size	Withdrawal quantity
50 Kg	500 g/h
65 Kg	650 g/h
500 Kg	5 Kg/h
1000 Kg	10 Kg/h

In case of higher chlorine requirement several chlorine containers of the same temperature have to be connected with header lines.

Withdrawal of liquid chlorine

- Within chlorination plants only possible from chlorine drums.
- Application of an evaporator is necessary.

Caution

Do not completely evacuate the chlorine drums. Danger of withdrawing deposits!

- Withdrawal of gaseous chlorine: Observe residual pressure of ca. 2 bar.
- Withdrawal of liquid chlorine: Observe residual pressure of ca. 4 bar.

After the extraction

- Close container valve.
- Disconnect container from the plant.
- Screw on valve protection nuts.
- Screw on protection cap.

Caution

Immediately close connection lines. No moisture must get into the lines!

2.3 Checking the tightness

Before commissioning check the tightness of the **whole plant**.



Warning

Do not check the tightness until the whole plant is ready for start-up.
Danger of chlorine break-out!

2.3.1 Checking the chlorine solution lines and the diaphragm non-return of the injector

- Observe the manual of the injector!

2.3.2 Checking the tightness of the vacuum lines

Vacuum lines are all lines between vacuum regulator and injector.

- Close all container valves
- Close the rate valve
- Open the shut-off valve at the injection unit
- Open motive water valve
- Switch on the booster pump
- Open the rate valve
 - Floater shows gas flow or vacuummeter shows more than - 9 m w. c.: **Leakage in vacuum line!**
- Close rate valve
- Switch off booster pump
- Close motive water valve
- Close the shut-off valve at the injection unit
- Check the vacuum lines and the connection. If necessary, carefully re-tighten them.
- Check tightness again!**
 - Floater shows no gas flow, vacuummeter shows -9 m w. c. or less: **Vacuum lines are tight.**

More possible reasons for insufficient operating vacuum

Note

- injector layed out too weakly or defective
- injector obstructed
- booster pump layed out too weakly or defective

2.3.3 Checking the tightness of pressure gas lines

Pressure gas lines are all lines from the gas containers to the vacuum regulator.

- If the plant is equipped with a nitrogen rinsing device: Check tightness roughly with nitrogen
- Detailed checking with ammonia

Checking the tightness with nitrogen

- Close all container valves
- Open container connection valves and all shut-off valves up to the gas dosing system
- Open the connection valve of the nitrogen cylinder
- Slowly open the valve of the nitrogen cylinder, until the lines have a pressure of about 10 bar (read at the manometer of the vacuum regulator).



Warning

Maximum nitrogen pressure 16 bar!
Danger of damages and gas leakage when being exceeded!

- Apply soap water to all components under pressure
 - Formation of bubbles and/or pressure drop at manometer
- Leakage in pressure lines!**
- Depressurize the plant!**
- Eliminate leakage!**
- Check tightness again!**

No formation of bubbles, pressure at manometer does not drop significantly within one hour: **Pressure lines are tight.**

Checking the tightness with ammonia

See chapter [5.2.2 Checking the tightness](#).

2.4 Constructional requirements of chlorination plants

All chlorine rooms



Fig. 6 Warning sign DIN 4844

- Warning sign according to DIN 4844 part 1 no. 4.2, to be installed at entrances

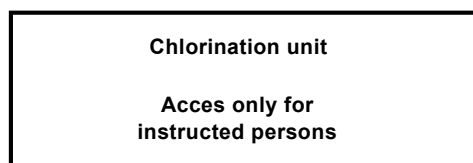


Fig. 7 Supplementary sign

- Supplementary sign according to DIN 4844 part 1 no. 4.5, to be installed at entrances



Fig. 8 Mandatory sign

- Mandatory sign according to DIN 4844 part 1 no. 4.3, to be installed inside the rooms

First aid in the case of CHLORINE GAS POISONING	
HEALTH DANGERS Chlorine gas is highly toxic and can cause severe damage to the respiratory system, eyes, and skin. It is a strong irritant and can lead to pulmonary edema and other life-threatening conditions. FIRST AID If exposed to chlorine gas, move to fresh air immediately. If inhaled, seek medical attention. If eyes are affected, flush with water. If on skin, remove contaminated clothing and wash the area with water.	

Fig. 9 Instruction sheet for first aid

- Instruction sheet for first aid in case of chlorine gas intoxications, to be installed inside the rooms
- Maximum temperature: 50 °C
 - recommended temperature: 18 - 20 °C
 - recommended minimal temperature: 15 °C
- Overpressure lines of dosing equipment must not end in the open air
- Chlorine rooms must not be dedicated for the permanent stay of people.
- Only chlorine containers and the chlorination plant may be present in the rooms.

Rooms with pressure lines (e. g. storage rooms for chlorine containers)

- Flat, even floor
 - not below ground level
 - not higher than a possible loading ramp

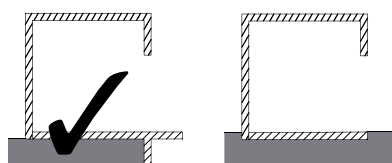


Fig. 10 Regulations for chlorine rooms (1)

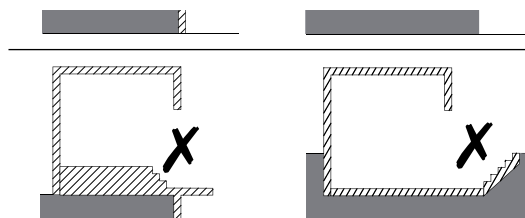
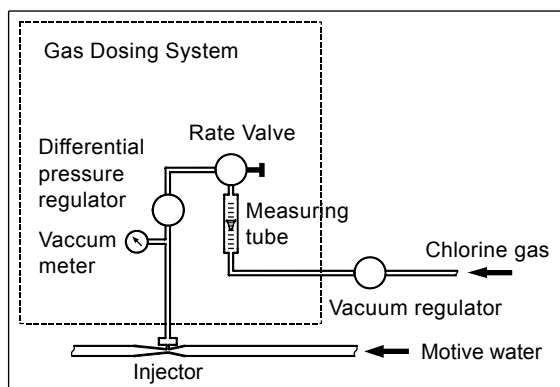


Fig. 11 Regulations for chlorine rooms (2)

- Direct exit to the open air
- Lockable
 - doors must open outwards
 - it must be possible to open the doors without a key from inside the room
- No connection to other rooms
 - separated gastight and fire-resistant from other rooms
- A maximum of two vent holes of max. 20 cm² each
- Water sprinkling system
 - for precipitating escaping chlorine gas
 - operation must be possible by hand from outside the chlorine rooms
 - sufficiently dimensioned run-off with air trap
- Chlorine gas warning system
 - with optical and acoustical alarm
 - coupled to the water sprinkling system
 - coupling must reactivate automatically after having switched off (e. g. for container exchange)
- Chlorine gas must not be able to get into lower-lying rooms, shafts, pits, canals or aspirating holes of ventilation systems.

2.5 Principle function of the components



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Fig. 12 Components of a chlorinator

2.5.1 Vacuum regulator

- Reduces the gas pressure (overpressure) to subatmospheric pressure (vacuum)

2.5.2 Measuring tube

- Displays the gas flow

2.5.3 Rate valve

- For adjustment of the required gas flow
 - manually or automatically

2.5.4 Differential pressure regulator (Option: VGA-117)

- Regulates the difference of the pressures before and after the rate valve to a constant value
 - Adjusted dosing flow stays constant even when the injector vacuum varies.

2.5.5 Vacuummeter (Option: VGA-117)

- Displays the injector vacuum

2.5.6 Injector

- Creates the vacuum necessary for operating the pant
- Mixes the chlorine gas with the water

2.6 List of valid laws and regulations

Laws and Regulations

BGBI. I 1975 S. 729	Verordnung über Arbeitsstätten (Arbeitsstättenverordnung - ArbStättV) Regulations concerning places of work (ArbStättV)
BGBI I 1980 S. 173, 184	Verordnung über Druckbehälter, Druckgasbehälter und Füllanlagen (Druckbehälterverordnung - DruckbehV) mit technischen Regeln Druckbehälter (TRB) mit technischen Regeln Druckgase (TRG) Regulations concerning pressure containers, gas cylinders and filling systems (pressure containers - DruckbehV) with technical rules for pressure containers (TRB) with technical rules for pressure gases (TRG)
BGBI I 1986 S. 1470	Gefahrstoffverordnung (GefStoffV) Regulations for hazardous materials (GefStoffV)
BGBI I 1975 S. 2494	Verordnung über gefährliche Arbeitsstoffe (Arbeitsstoffverordnung - ArbStoffV) Regulations for hazardous work materials (ArbStoffV)
GUV 0.1	Unfallverhütungsvorschrift "Allgemeine Vorschriften" Regulations for accident prevention "General Regulations"
GUV-V D5	Unfallverhütungsvorschrift "Chlorung von Wasser" Regulations for accident prevention "Chlorination of water"
GUV 49.1	Prüfliste zur Unfallverhütungsvorschrift "Chlorung von Wasser" Checklist for Regulations for accident prevention "Chlorination of water"
GUV 0.3	Unfallverhütungsvorschrift "Erste Hilfe" Regulations for accident prevention "First-Aid"
GUV 20.5	Merkblatt "Anleitung zur ersten Hilfe bei Unfällen" Leaflet "Instructions for First-Aid in case of accidents"
GUV 20.6	Merkblatt "Verbandzeug für die erste Hilfe bei Unfällen" Leaflet "Dressing material for First-Aid in case of accidents"
GUV 20.14	Atemschutzmerkblatt Leaflet for the protection of the air
GUV 29.6	Merkblatt "Über den Umgang mit ätzenden Stoffen" Leaflet "Handling of caustic materials"
GUV 60.3G26	Grundsatz für arbeitsmedizinische Vorsorgeuntersuchungen "Träger von Atemschutzgeräten für Arbeit und Rettung" G 26 Principle for preventive industrial medicine checkup "Wearers of respiratory equipment for work and rescue" G 26
GUV 2.6	Unfallverhütungsvorschrift "Druckbehälter" Regulations for accident prevention "Pressure containers"
GUV 2.10	Unfallverhütungsvorschrift "Elektrische Anlagen und Betriebsmittel" Regulations for accident prevention "Electrical installations and resources"
GUV 9.9	Unfallverhütungsvorschrift "Gase" Regulations for accident prevention "Gases"
ZH1/230	Merkblatt "Chlor" Leaflet "Chlorine"
CEFIC	Unfallmerkblatt für den Straßentransport "Chlor" Klasse 2, Ziffer 3 at UN 2201 Accident leaflet for the road transport "Chlorine" Class 2, Number 3 at UN 2201
BGBI I 1985, S. 1550	Verordnung über die Beförderung gefährlicher Güter auf der Straße - Gefahrgut Vstr/GGVS - Klasse 2, Ziffer 3 at Regulations for the transport of dangerous goods on the road - Gefahrgut Vstr/GGVS - Class 2, Number 3 at
GGVE	Gefahrgutverordnung Eisenbahn, Klasse 2, Ziffer 3 at) Regulations for dangerous goods on trains, Class 2, Number 3 at)
RID	Internationale Verordnung für die Beförderung gefährlicher Güter mit der Eisenbahn - Klasse 2, Ziffer 3 at International regulations for the transport of dangerous goods by train - Class 2, Number 3 at
BGBI I 1977, S. 1119	Verordnung über die Beförderung gefährlicher Güter auf dem Rhein (ADNR) Regulations for the transport of dangerous goods on the Rhine (ADNR)

DVGW-Rules

W203	Begriffe der Chlorung Concept of chlorination
W645-1	Überwachungs-, Mess-, Steuer- und Regeleinrichtungen in Wasserversorgungsanlagen Devices for monitoring, measuring, control and regulation in water supply plants
W291	Desinfektion von Wasserversorgungsanlagen Disinfection of water supply plants
W623	Dosieranlagen für Desinfektions- bzw. Oxidationsmittel - Dosieranlagen für Chlor Dosing units for disinfection or oxidation with chlorine
W640	Überwachungs-, Meß-, Steuer- und Regeleinrichtungen in Wasserwerken Systems for monitoring, measurement, control and regulation in waterworks
	DVGW-Merkblatt Arbeitshilfe zur Erstellung einer örtlichen Betriebsanweisung für Chlorungsanlagen unter Verwendung von Chlorgas DVGW leaflet Aid for the creation of a local manual für chlorination systems using chlorine gas

Standards

DIN 19606	Chlorgasdosieranlagen zur Wasseraufbereitung Chlorine gas dosing systems for water treatment
DIN 19607	Chlor zur Wasseraufbereitung Chlorine for water treatment
DIN EN 937	Chlor zur Aufbereitung von Wasser für den menschlichen Gebrauch Chlorine for the treatment of water for the human use
DIN 19643	Aufbereitung von Schwimm- und Badewasser Treatment of swimming pool and bathing water
DIN 3179, Teil 1,2	Einteilung der Atemgeräte, Übersicht Division of the respiratory equipment, overview
DIN 4102, Teil 2	Brandverhalten von Baustoffen und Bauteilen Behaviour in fire of building materials and parts
DIN 477, Teil 1	Gasflaschenventile; Bauformen, Baumaße, Anschlüsse, Gewinde Gas cylinder valves; forms, measurements, connections, threads

2.7 Recommended diameter**2.7.1 Between vacuum regulator and dosing regulator**

Length of the vacuum line in (m)	Dosing quantity (g/h)						
	40	100	250	500	1000	2000	4000
0	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10
10	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10
20	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15
30	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15
40	DN 8	DN 8	DN 8	DN 8	DN 8	DN 15	DN 15
50	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15	DN 15
75	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15	DN 15
100	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15	DN 20

Grundfos Alldos company standard calculated with pressure drop

p=12.5 mbar

The indications in this table result from pressure loss observations. They do not take into consideration the possible influence of length and diameter of lines on the operational reliability of the system.

2.7.2 Between dosing regulator and injector

Length of the vacuum line in (m)	Dosing quantity (g/h)						
	40	100	250	500	1000	2000	4000
0	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8
10	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8
20	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10
30	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10
40	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10
50	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15
75	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15
100	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15

Grundfos Alldos company standard calculated with pressure drop

p=50 mbar

The indications in this table result from pressure loss observations. They do not take into consideration the possible influence of length and diameter of lines on the operational reliability of the system.

3. Technical Data

3.1 Type key dosing regulator VGA-113

Example: Gas dosing regulator Type key VGA-113-100/M0, Y

Code	Example	VGA-113	-100	/M	0	-N
	Vaccuperm Gas Advanced = VGA					
Dosing flow						
25	1-25 g/h					
40	2-40 g/h					
100	5-100 g/h					
250	10-250 g/h					
500	25-500 g/h					
1000	50-1000 g/h					
2000	100-2000 g/h					
3000	150-3000 g/h					
4000	400-4000 g/h					
Adjustment						
M	Manual					
A	Automatic, 230-240 V / 50-60 Hz					
B	Automatic, 110-115 V / 50-60 Hz					
Control						
0	Without					
1	4-20 mA					
2	Direct					
Differential pressure regulator						
Y	Yes, with vaccumeter					
N	No					

3.2 General Data

Permissible medium	Cl ₂
Accuracy	± 4 % of upper limit
Control range	1:20
Flow meter	according to the floater principle, length of the measuring tube 70 mm
Weight	0.9 kg
Vacuum connection (from the vacuum regulator)	for hose 8/11
Vacuum connection (to the injector)	for hose 8/11
Weight without Servomotor	0.9 kg
Weight with Servomotor	3.1 kg
Permissible vacuum regulator	e.g. VGA-111

3.2.1 Dosing flow

1.5 - 25 g/h	0.07 - 1.3 lbs/day
2 - 40 g/h	0.1 - 2 lbs/day
5 - 100 g/h	0.2 - 5 lbs/day
10 - 250 g/h	0.5 - 14 lbs/day
25 - 500 g/h	1.5 - 26 lbs/day
50 - 1000 g/h	2.5 - 50 lbs/day
100 - 2000 g/h	5 - 100 lbs/day
150 - 3000 g/h	10 - 150 lbs/day
400 - 4000 g/h	30 - 220 lbs/day

Possible total dosing flow for several dosing regulators: max. 4000 g/h

3.2.2 Accessories (not including)

Installation material: Hoses of different lengths
Holding plate for wall fixing at change of container
Test medium for leak search

3.3 Dimensioned drawings

3.3.1 Dosing regulator

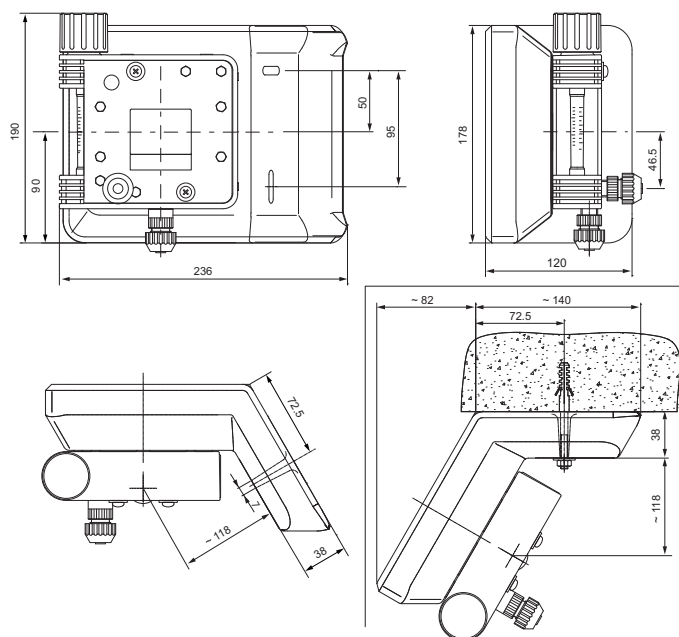


Fig. 13 Dosing regulator

3.3.2 Dosing regulator with servomotor

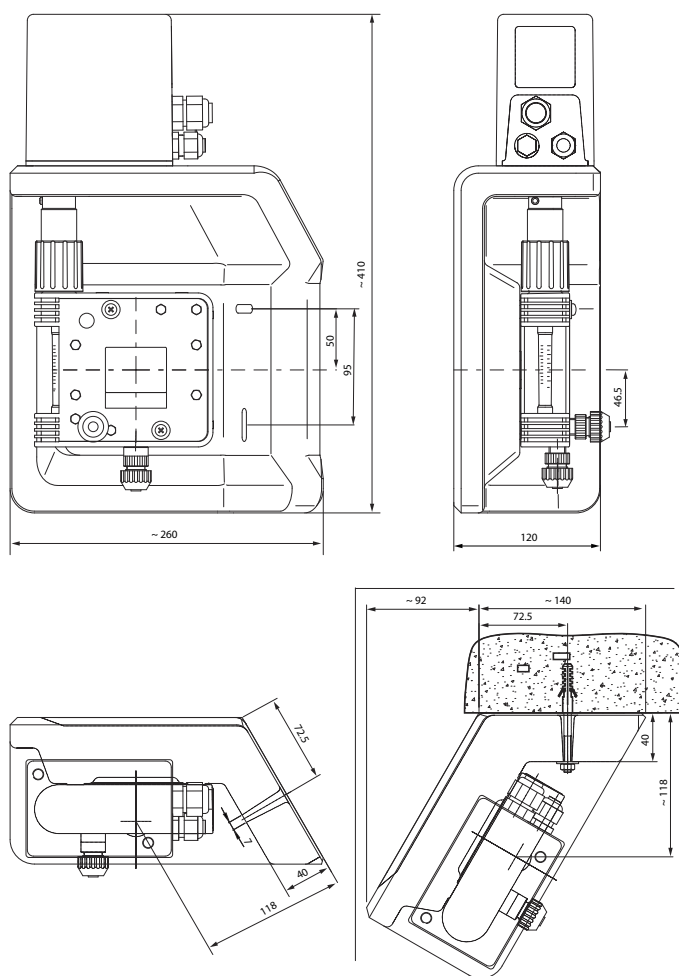


Fig. 14 Dosing regulator with servomotor

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3.4 Electrical data for VGA-113

3.4.1 Servomotor with current input

Servomotor	Synchronous motor with limit switches
Current input	4 - 20 mA
Actuating time 0-100 %	ca.95 s (50 Hz), ca. 76 s (60 Hz)
Power consumption	3 VA
Degree of protection	IP55
Permissible ambient temperature	0 °C to +50 °C

Type key	Voltage	Nominal current
VGA-113-XXX/A1, N	230 V (50/60 Hz)	0.011 A
VGA-113-XXX/B1, N	115 V (50/60 Hz)	0.022 A

3.4.2 Servomotor with feedback potentiometer

Servomotor	Synchronous motor with limit switches
Feedback potentiometer	Single potentiometer
Actuating time 0-100 %	ca.95 s (50 Hz), ca. 76 s (60 Hz)
Power consumption	3 VA
Degree of protection	IP55
Permissible ambient temperature	0 °C to +50 °C

Type key	Voltage	Nominal current
VGA-113-XXX/A2, N	230 V (50/60 Hz)	0.011 A
VGA-113-XXX/B2, N	115 V (50/60 Hz)	0.022 A

4. Installation

4.1 Transport and storage

- Handle with care, don't throw!
- Dry and cool storage place.

4.2 Unpacking

- Observe when unpacking:
 - No humidity should get into gas-leading parts!
 - No foreign matter should get into gas-leading parts!
- Mount as soon as possible after unpacking.

4.3 Typical installations

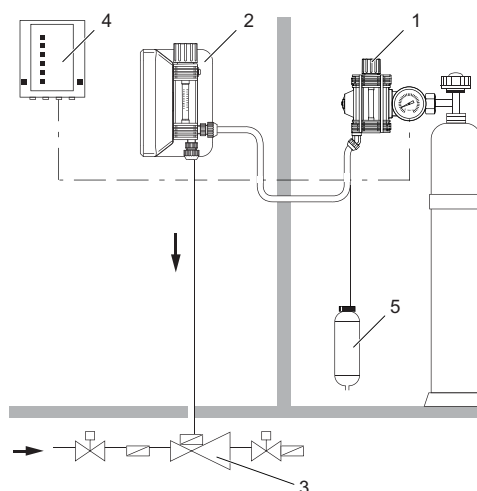


Fig. 15 Vacuum regulator and dosing regulator

Item	Description
1	Vacuum regulator e.g. VGA-111 (with contact pressure gauge)
2	Dosing regulator VGA-113
3	Injector
5	Remote display for container empty indication
6	Adsorption device

4.4 Mounting



Warning

Observe the information, see chapter 2. *Handling chlorine*

4.4.1 Dosing regulator

- Mount the dosing regulator to the wall.

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5. Commissioning

5.1 Preparations for commissioning

5.1.1 Vacuum connections



Warning

Before connecting, ensure that the valves of all gas containers are closed.

Only use the intended line types!

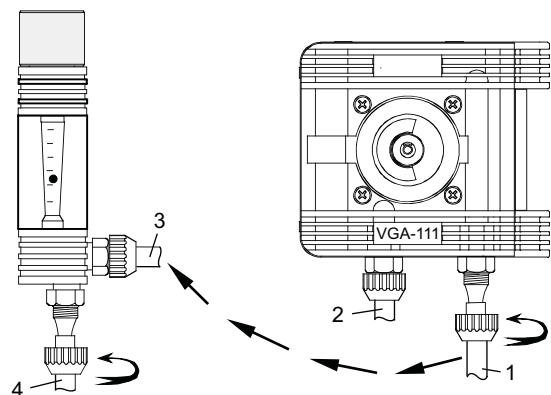


Fig. 16 Vacuum connections

Item	Description
1	Vacuum line to the vacuum regulator
2	Overpressure line
3	Vacuum line to the dosing regulator
4	Vacuum line to the injector

Only tighten the union nuts of the vacuum connections by hand. Do not use any tools!

Caution

Danger of damages!

Maximum length of the vacuum line, see chapter [2.7 Recommended diameter](#)

- At the vacuum regulator, connect the vacuum line (1) and connect it to the dosing regulator (3).

Warning

Lead the overpressure line downwards, as chlorine is heavier than air.

Connect the overpressure line to a suitable adsorption device. If using a gas warning device: Mount the sensor in a distance of about 10 cm from the outlet hole of the adsorption device.

- Connect the overpressure line (2).
- Connect the vacuum line to the injector (4).

5.1.2 Electrical connections

Before connecting the mains cable:

Check if the mains voltage indicated on the type plate corresponds to the local mains voltage. Wrong voltage can lead to damages of the product!

Caution

To ensure electromagnetic compatibility (EMC): Signal lines must be shielded. The shield must be connected to PE or ground on the side of the counterpart, par exemple a controller.

Signal lines and mains lines must be guided in different cable channels.

Vacuum regulator e.g. VGA-111 with liquid trap and/or contact pressure gauge

- Connect the heating of the optional liquid trap.
 - Connect the mains line to the mains.
- Connect the optional contact manometer.
 - Connect two cables with an external evaluation device.

Note

The polarity is arbitrary.

Dosing regulator VGA-113 with servomotor

Warning

Disconnect from mains before removing the cover!

Observe local safety precautions!

Protect cable connections and plugs against corrosion and moisture.

Close unused screwings with suitable caps!

- Establish the electrical connections according to the following terminal diagrams.
 - with current input 4-20 mA

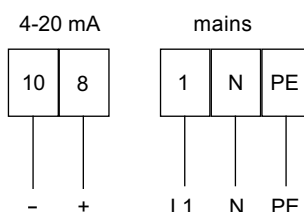


Fig. 17 Terminal connections, current input

– without current input, with feedback potentiometer

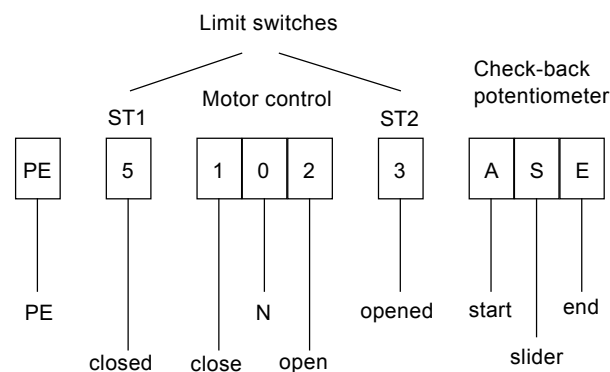


Fig. 18 Terminal connections, with cheedback potentiometer



Warning

The terminals 3 and 5 of the limit switches are on mains potential when the switch closes after the limit position had been reached.

5.2 Checks before commissioning

Check tightness of the total plant before start-up.



Warning

Check the tightness not until the total plant is ready for start-up.

Danger of gas leakage!

5.2.1 Checking the gas solution lines and the non-return diaphragm of the injector

– Observe the manual of the injector!

5.2.2 Checking the tightness



Warning

Liquid ammonia must not come in contact with parts of the plant!

Danger of leakages by corrosion!



Fig. 19 No liquid ammonia

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5.2.3 Checking the pressure connections (after changing gas cylinder)

- Open the cylinder valve and quickly close it again.
- Slowly pass the open ammonia bottle along gas-leading parts.
 - Slightly press the bottle in a pumping manner, allowing the ammonia mist to rise up.
 - Formation of white mist: Leakage at the pressure connection!
- Depressurize the plant!
- Eliminate leakage!
- Check tightness again

Formation of white mist --> pressure connection is not tight

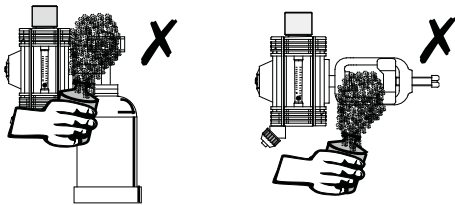


Fig. 20 Ammonia + chlorine gas --> formation of white mist

No formation of white mist --> pressure connection is tight

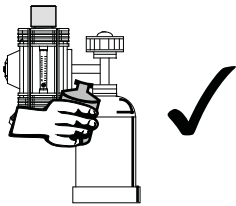


Fig. 21 No chlorine gas --> no formation of white mist

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Checking the inlet valve

- Remove the union nut at the overpressure connection and take off the overpressure line.
- Close the rate valve.
- Open the cylinder valve.
- Slowly pass the open ammonia bottle along the overpressure connection.
 - Slightly press the bottle in a pumping manner, allowing the ammonia mist to rise up.
 - Formation of white mist: **The inlet valve is not tight!**
- **Depressurize the plant!**
- **Check the inlet valve and repair it!**
- **Check tightness again**

Formation of white mist --> the inlet valve is not tight!

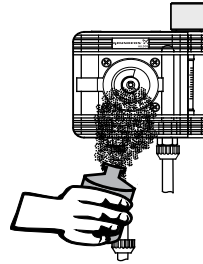


Fig. 22 Ammonia + chlorine gas --> formation of white mist

No formation of white mist --> the inlet valve is tight!

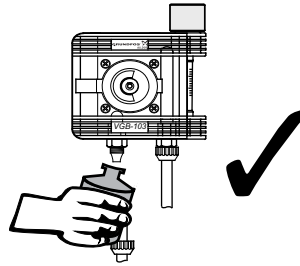


Fig. 23 No chlorine gas --> no formation of white mist.

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6. Operationh

6.1 Description of the device

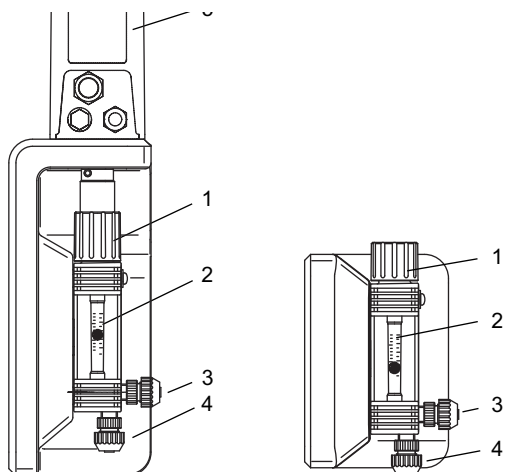


Fig. 24 Dosing regulator with servomotor and manual dosing regulator

Item	Component	Description
1	Rate valve	For manual adjustment of the desired gas flow.
2	Measuring tube	It displays the actual gas flow.
3	Connection for the vacuum line from the vacuum regulator	For the injection point
4	Connection for the vacuum line to the injector	For the injection point
	optional: Differential pressure regulator (not visible)	It keeps the difference of the pressures before and after the rate valve on a constant value. The adjusted dosing flow stays constant even when the injector vacuum varies.
5	Only for VGA-113 with servomotor	For automatic adjustment of the desired gas flow.

Caution The rate valve is not a perfectly closing shut-off valve!

6.2 Automatic adjustment of the dosing flow

Only with option servomotor!

Note In case of automatic adjustment of the dosing flow, no adjustment can be made at the dosing unit. All adjustment has to be made at the control device. Observe the manual of the control device!

6.2.1 Switching from automatic to manual adjustment of the dosing flow; only with option servomotor

- Switch off automatic control.
 - Push the coupling (1) upwards.
 - Move the retention pin (2) through the coupling, until it cannot move down any longer.
 - Now the dosing flow can be adjusted by hand.
- > The servomotor is disconnected from the rate valve.

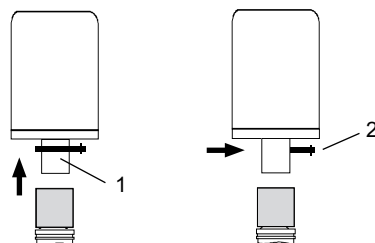


Fig. 25 Disconnect to servomotor

6.2.2 Switching from manual to automatic adjustment of the dosing flow; only with option servomotor

- Move back the retention pin (2).
 - Now the coupling (1) falls down.
 - Turn the adjustment knob (3), until the coupling snaps in.
- > Now the servomotor is connected to the rate valve.

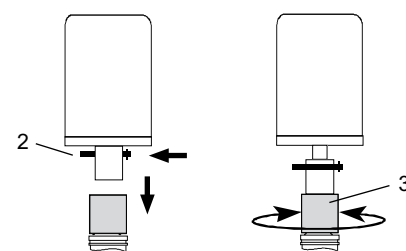


Fig. 26 Connect to servomotor

6.3 Adjusting the Dosing Flow

6.3.1 Increasing the dosing flow

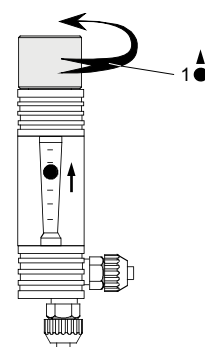


Fig. 27 Increasing the dosing flow

- Turn the adjustment knob of the rate valve **slowly** to the right.

Caution Do not screw out the adjustment knob. Do not continue turning the knob when maximum dosing flow is reached.

6.3.2 Decreasing the dosing flow

- Turn the adjustment knob of the rate valve slowly to the left.

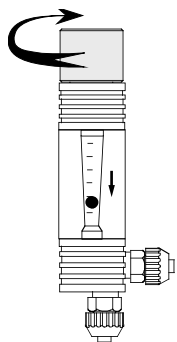


Fig. 28 Decreasing the dosing flow

6.4 Reading the dosing flow

- Read the dosing flow at the upper edge of the ball.

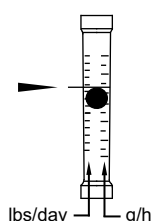


Fig. 29 Reading the dosing flow

Note

The scale of the measuring tube is adjusted to a gas temperature of 20 °C. In case of extremely different temperatures the dosing flow may differ from the value being displayed.

6.5 Switching on

- Close the rate valve.
- Open the shut-off valve at the injection unit.
- Open the motive water valves.
- Switch on the motive water of the injector.
- Open the valve of the gas cylinder.
- Slowly open the rate valve until the ball in the measuring tube shows the desired gas flow.

6.6 Switching off

6.6.1 Emergency stop

In case of gas break-out



Warning

Immediately leave the room in case of gas break-out and put on safety equipment!

Then start counter-measures!

- Immediately leave the room!
- Start counter-measures according to local safety precautions.

In case of small leakages

- Put on safety equipment!
- Immediately close the cylinder valve.
- Let the system run until all parts are evacuated.
- Switch off the system as described in the following.



Warning

Repairs of components of the system only by authorized personnel!

6.6.2 Switching off while the system is running

- Close the cylinder valve.
- Let the system run until the measuring tube shows no more gas flow.
- Close the rate valve.
- Switch off the motive water of the injector.
- Close the motive water valves.
- Close the shut-off valve at the injection unit.

6.7 Changing the gas cylinder

The empty indication of the dosing unit shows when the minimum pressure of the gas cylinder is reached. Then the cylinder must be changed.



Warning

Put on the prescribed safety equipment!

While the system is running

- Close the cylinder valve.
- Let the system run until the measuring tube shows no more gas flow.
- Close the rate valve at the dosing regulator.
- Switch off the motive water of the injector.
- Close the motive water valves.
- Close the shut-off valve at the injection unit.
- Unscrew the vacuum regulator from the connection of the gas cylinder.
 - If present, place it on the wall holding device.
- Screw the protective cap onto the empty cylinder, move it away and protect it against falling down.
- Place the new gas cylinder and fix it correctly.
- Remove the protective cap from the new gas cylinder.
 - All connections must be absolutely clean and dry.



Warning

Never re-use used gaskets! Replace the gaskets at any change of gas container!

Never use the vacuum regulator without the filter!

- Insert the gasket at the pressure connection of the vacuum regulator.
 - Replace the filter, if necessary.
- Screw the vacuum regulator onto the connection of the cylinder.

Caution

Do not align the device by turning the housing!

If necessary, hold up with a fork wrench size 13!



Warning

Before re-starting: check the tightness!

- Check the tightness.
- Re-start the system.

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6.8 Possible faults

Fault	Cause	Correction
Gas leakage at the pressure connection.	The gasket at the pressure connection is damaged.	Replace the gasket.
	Sealing surfaces are dirty, there are rests of old gaskets on the sealing surfaces.	Clean the sealing surfaces.
	The sealing surface at the connection of the gas cylinder is damaged.	Use a different gas cylinder.
	The pressure connection is loose.	Tighten the pressure connection.
	The sealing surface at the pressure connection of the device is damaged.	Call the service
Gas escapes at the overpressure line.	Re-liquefacted gas enters the inlet valve.	Reduce the extraction amount, ensure correct course of temperature.
	The inlet valve is soiled or damaged.	Call the service
Formation of ice on the gas cylinder.	The extraction amount per hours is higher than 1 % of the contents of the gas cylinder, the system is too small.	Use a dosing system with more than one gas cylinders.
	The extraction amount per hours is higher than 1 % of the contents of the gas cylinder, the system is too small.	Call the service
Desired dosing flow is not reached.	Insufficient injector vacuum.	Check the injector, repair it if necessary. Observe the manual of the injector.
		Check the motive water pump, repair it if necessary.
		Check the dirt trap before the injector, clean it if necessary.
	Leakage in vacuum line between dosing unit and injector.	Eliminate the leakage.
	Vacuum line between dosing unit and injector is too long.	Observe table "Suggested line lengths and diameters".
	Vacuum line is dirty.	Replace the vacuum line.
	The valve of the gas cylinder is closed.	Open the valve of the gas cylinder.
	The gas cylinder is empty.	Replace the empty gas cylinder by a new one.
	Filter at the pressure connection of the dosing unit is soiled.	Call the service
Despite correctly displayed gas flow the desired gas solution concentration cannot be reached.	Wrong adjusting spindle (diameter too low).	Call the service
	Leakage in vacuum line between vacuum regulator and dosing unit.	Eliminate the leakage.
	Servomotor is switched to manual operation.	Switch servomotor to automatic operation.
	The ball in the measuring tube is caught.	Call the service
	Non-return diaphragm of the injector is defective.	Call the service
	The servomotor or the control device is adjusted wrongly.	Call the service
	The servomotor or the control device is adjusted wrongly.	Call the service
	The servomotor or the control device is adjusted wrongly.	Call the service
	The servomotor or the control device is adjusted wrongly.	Call the service

7. Maintenance

Rates for cleaning and maintenance

- at least every 12 months
- in case of malfunction



Warning

Switch off the whole plant before doing cleaning and maintenance work!

Danger of gas break-out!



Warning

Before re-starting, check the tightness!

Danger of gas break-out!

Only for execution with limiting nozzle:

Execute service for all vacuum regulators of a container battery at the same time!

8. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Declaration of conformity

GB: EU declaration of conformity

We, Grundfos, declare under our sole responsibility that the products VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148, to which the declaration below relates, are in conformity with the Council Directives listed below on the approximation of the laws of the EU member states.

DE: EU-Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass die Produkte VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148, auf die sich diese Erklärung beziehen, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EU-Mitgliedsstaaten übereinstimmen.

FR: Déclaration de conformité UE

Nous, Grundfos, déclarons sous notre seule responsabilité, que les produits VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148, auxquels se réfère cette déclaration, sont conformes aux Directives du Conseil concernant le rapprochement des législations des États membres UE relatives aux normes énoncées ci-dessous.

PL: Deklaracja zgodności UE

My, Grundfos, oświadczamy z pełną odpowiedzialnością, że nasze produkty VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148, których deklaracja niniejsza dotyczy, są zgodne z następującymi dyrektywami Rady w sprawie zbliżenia przepisów prawnych państw członkowskich.

RO: Declarația de conformitate UE

Noi Grundfos declarăm pe propria răspundere că produsele VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148, la care se referă această declarație, sunt în conformitate cu Directivele de Consiliu specificate mai jos privind armonizarea legilor statelor membre UE.

SK: ES vyhlásenie o zhode

My, spoločnosť Grundfos, vyhlasujeme na svoju plnú zodpovednosť, že produkty VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148 na ktoré sa vyhlásenie uvedené nižšie vzťahuje, sú v súlade s ustanoveniami nižšie uvedených smerníc Rady pre zbliženie právnych predpisov členských štátov EÚ.

CZ: Prohlášení o shodě EU

My firma Grundfos prohlašujeme na svou plnou odpovědnost, že výrobky VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148, na které se toto prohlášení vztahuje, jsou v souladu s níže uvedenými ustanoveními směrnice Rady pro sblížení právních předpisů členských států Evropského společenství.

ES: Declaración de conformidad de la UE

Grundfos declara, bajo su exclusiva responsabilidad, que los productos VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148 a los que hace referencia la siguiente declaración cumplen lo establecido por las siguientes Directivas del Consejo sobre la aproximación de las legislaciones de los Estados miembros de la UE.

HU: EU megfelelősségi nyilatkozat

Mi, a Grundfos vállalat, teljes felelősséggel kijelentjük, hogy a(z) VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148 termékek, amelyre az alábbi nyilatkozat vonatkozik, megfelelnek az Európai Unió tagállamainak jogi irányelveit összehangoló tanács alábbi előírásainak.

PT: Declaração de conformidade UE

A Grundfos declara sob sua única responsabilidade que os produtos VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148, aos quais diz respeito a declaração abaixo, estão em conformidade com as Directivas do Conselho sobre a aproximação das legislações dos Estados Membros da UE.

RU: Декларация о соответствии нормам ЕС

Мы, компания Grundfos, со всей ответственностью заявляем, что изделия VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148, к которым относится нижеприведённая декларация, соответствуют нижеприведённым Директивам Совета Евросоюза о тождественности законов стран-членов ЕС.

TR: AB uygunluk bildirgesi

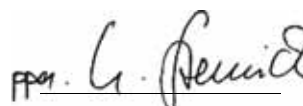
Grundfos olarak, aşağıdaki bildirim konusu olan VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148 ürünlerinin, AB Üye ülkelerinin direktiflerinin yakınlştırılmasıyla ilgili durumun aşağıdaki Konsey Direktifleriyle uyumlu olduğunu ve bununla ilgili olarak tüm sorumluluğun bize ait olduğunu beyan ederiz.

- Machinery Directive (2006/42/EC). Standards used: DIN EN ISO 12100:2010.
- Low Voltage Directive (2014/35/EU). Standards used: EN 60204-1:2007
EN 61010-1:2011-07*
- EMC Directive (2014/30/EU)*. Standards used: EN 55014-1:2006+A1:2009+A2:2011
EN 55014-2:2016
EN 61000-6-1:2007
EN 61000-6-3:2011
- Other regulations applied: DIN 19606:2010-09
- RoHS Directives (2011/65/EU and 2015/863/EU). Standard used: EN 50581:2012

* Only valid for products with electrical components

This EU declaration of conformity is only valid when published as part of the Grundfos installation and operating instructions (publication numbers 95714202, 95714224, 98028175, 95714246, 95714278, 95713929, 95714262, 95714296).

Pfinztal, 1st March 2018



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Установки вакуумные для дозирования газов типа Vassiperm сертифицированы на соответствие требованиям Технических регламентов Таможенного союза: ТР ТС 004/2011 "О безопасности низковольтного оборудования"; ТР ТС 010/2011 "О безопасности машин и оборудования"; ТР ТС 020/2011 "Электромагнитная совместимость технических средств".

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ТС N RU Д-DK.АИ30.В.05343, срок действия до 08.10.2020 г.

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